Chapter Name

National Income

Accounting

→ Difference between Micro and Macro economics.

Micro Economics	Macro Economics
1 It is study of individual economic unit of an economy.	It is the study of economy as a whole and its aggregates.
2 It deals with individual income, individual prices & individual output etc.	It deals with national income, general price rural and national output etc.
3 Its central problem is price determination and allocation of resources.	It central problem is determination of rural income.
4 Its main roots are demand and supply of particular commodity / factor.	Its main roots are aggregate demand and aggregate supply of economy as a whole.
5 It discusses how equilibrium of a consumer a producer or an industry is attained	It is concerned with determination of equilibrium level of income and employment.

→ Difference between Transfer income and Factor income.

	Transfer Income	Factor Income
1.	It is the income obtained without providing any factor input.	It is the income earned by providing factor input.
2.	It is a one-sided income i.e unilateral concept.	It is two-sided income i.e bilateral concept.
3.	It is an unearned income.	It is an earned income.
4.	It is not included in the calculation of national income.	It is included in the calculation of national income.
5.	Example :- Tax old-age pension	Example :- Bonus Pension

→ Examples of Transfer income

1. Money received by parents from their son working abroad.
2. Financial help is given by govt. to flood victims
3. Peize won in a lottery.
4. Tax
5. Old-age pension
6. Unemployment allowances.
7. Birthday gift received from a feriend.
8. Bonus received on Diwali.
9. Financial help of earthquake victims.

→ Examples of factor income

1. Salary received by a person from a company.
2. Compensation received from the employer.
3. Claim received from insurance company by an injured worker.
4. Bonus
5. Gift received from the employer.
6. Pension.

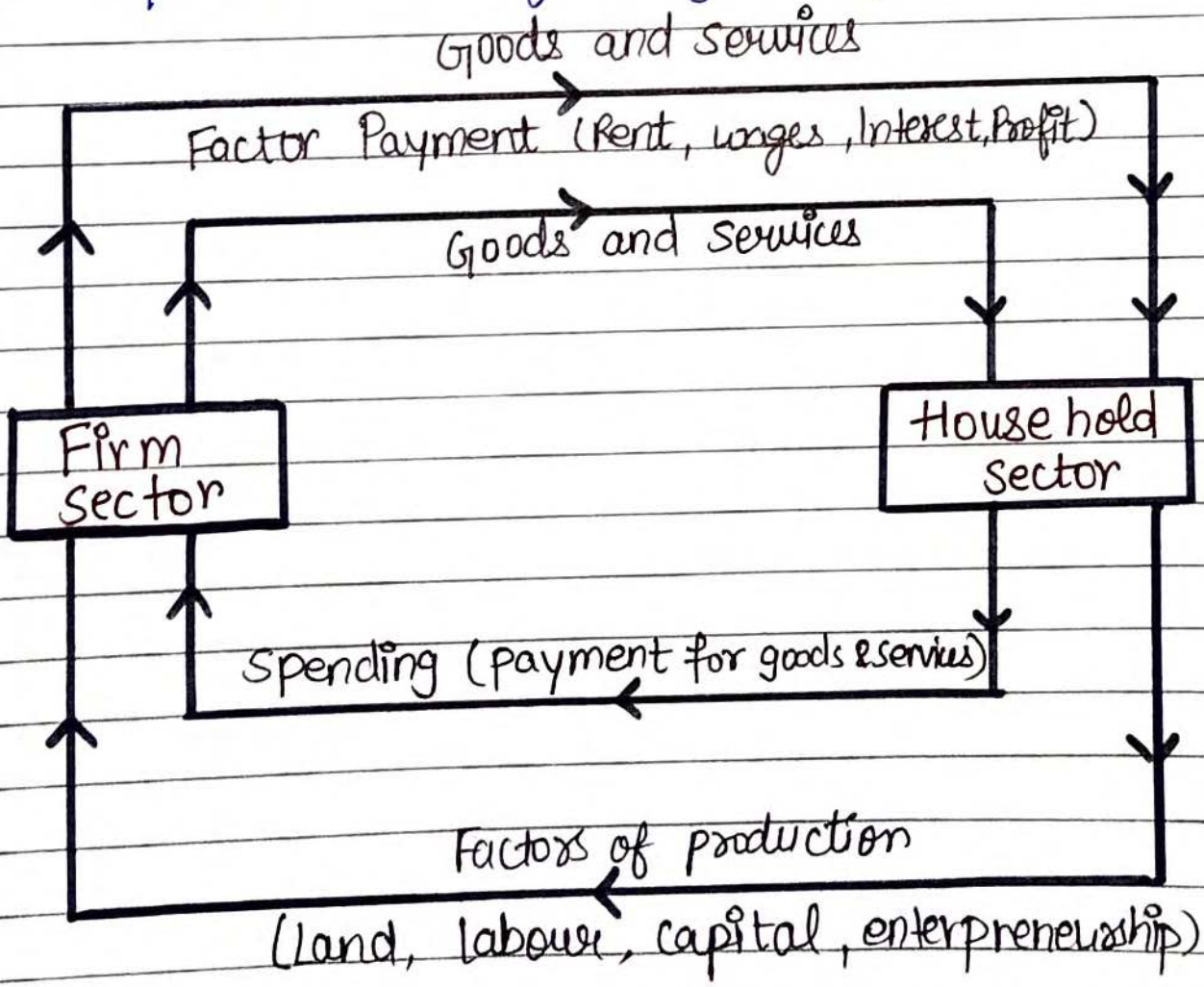
Circular flow of income under two-sector economy.

Circular flow of income refers to continual circular movement of money and goods in an economy.

It has three phases:

i) Income phase
ii) Expenditure phase
iii) Production phase

Circular flow of income under 2-sector economy is explained in following diagram's



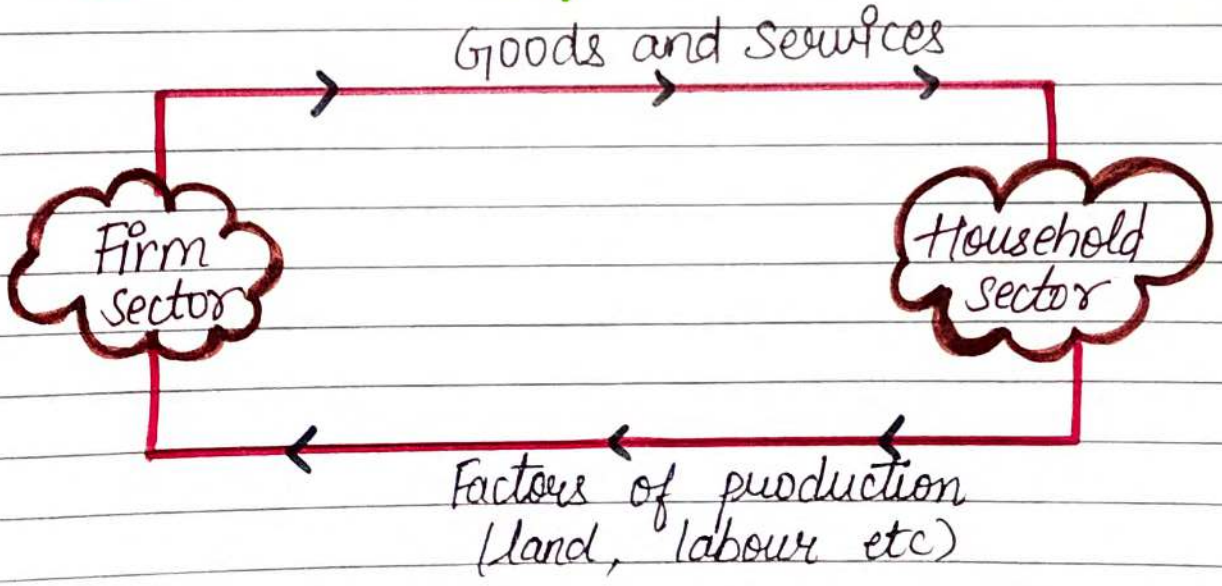
Initially, household sector provides factors of production to firm sector and in return firm sector gives factor payment to household sector.

Further, firm sector gives goods and services to household sector and in return household sector gives payment for goods of services to firm sector.

Types of Circular flow

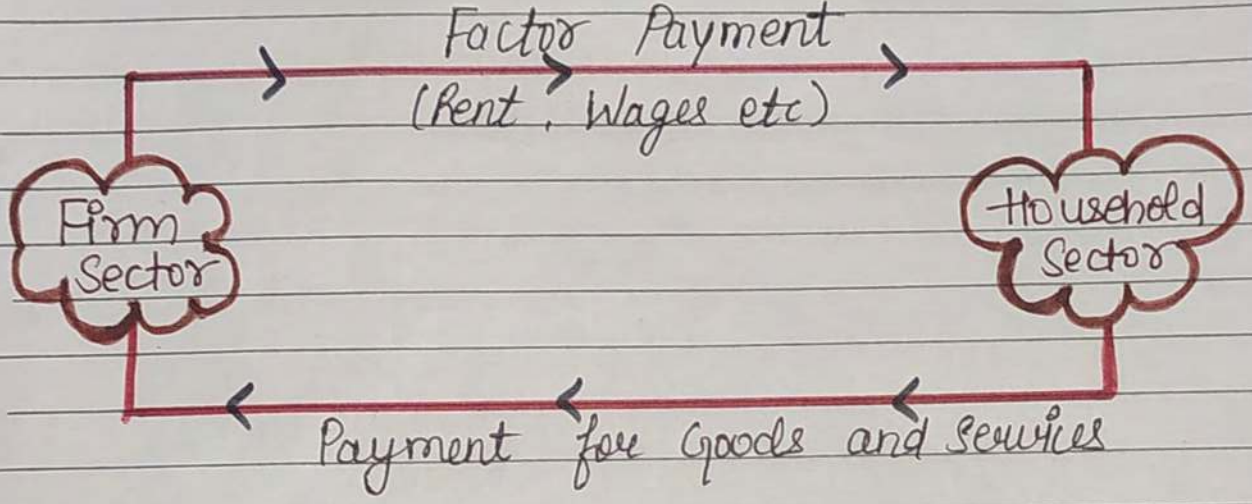
- Real flow
- Money flow

Real Flow :- It refers to the flow of non-inventory transactions in an economy.



Money flow :-

It refers to flow of monetary transactions in an economy. It is also known as nominal flow.



Related Aggregates

1. **Gross Domestic product at market price:**
It is the money value of all final goods and services produced during an accounting year with in the domestic territory of a country.

2. **Gross National product at market price:**
It is a money value of all final goods and services produced by a country during an accounting year including net factor income from abroad.

3. **Net factor income from abroad:**
Difference between the factor incomes earned by our residents from abroad and factor income earned by non-residents with in our country.

Components of Net factor income from abroad

1. Net compensation of employees
2. Net income from property and enterpreneurship (other than retained earnings of resident companies of abroad).
3. Net retained earnings of residents companies abroad.

Formulas

1. $NNP_{mp} = GNP_{mp}$ – depreciation
2. $NDP_{mp} = GDP_{mp}$ – depreciation
3. $NDP_{FC} = NDP_{mp}$ – Net Indirect taxes (indirect tax – Subsidies)
4. $GDP_{FC} = NDP_{FC}$ + depreciation
5. $NNP_{FC} = GDP_{mp}$ – depreciation + Net factor income from abroad – Net indirect taxes.
6. (NNP_{FC} is the sum total of factor income earned by normal residents of a country during the accounting year).
7. $NNP_{FC} = NDP_{FC}$ + Net factor income from abroad.

Concept of domestic territory

Domestic territory is a geographical territory administered by a government within which persons, goods and capital circulate freely. (Areas of operation generating domestic income, freedom of circulation of persons, goods and capital).

Scope identified as

1. Political frontiers including territorial waters and air space.
2. Embassies, consulates, military bases etc. located abroad but including those locates within the political frontiers.
3. Ships, aircrafts etc, operated by the residents between two or more countries.
4. Fishing vessels, oil and natural gas rigs etc. operated by the residents in the international waters or other areas over which the country enjoys the exclusive rights or jurisdiction.

Resident (normal resident)

Normal resident is a person or an institution who ordinarily resides in that country and whose center of economic interest lies in that country.
The Centre of economic interest implies :-

(1) The resident lives or is located within the economic territory.
(2) The resident carries out the basic economic activities of earnings, spending and accumulation from that location.
(3) His center of interest lies in that country.

Note :
(1) International organisations are not the resident of any country.
(2) People who work in International organisation are the resident of the country where they live.
(3) Workers from across the border are not the resident of the country where they work, they are resident of the country where they live.
(4) Foreign ships and Aircraft crew members are non-resident.
(5) Foreign visitors Travellers are resident of the country where they live.

Relation between national Product and domestic Product

Domestic product concept is based on the production units located within domestic (economic) territory, operated both by residents and non-residents. National product concept based on resident and includes their contribution to production both within and outside the economic territory.

National Product = Domestic product + Residents contribution to production outside the economic territory (factor income from abroad) – Non-residents contribution to production inside the economic territory (factor income to abroad).

Methods of Calculating National Income

- Product method
- Income method
- Expenditure method

1. Product Method (value added method)

i) Sales + change in stock = value of output
ii) Change in stock = closing stock − opening stock.
iii) Value of output − Intermediate consumption = GDP$_{mp}$
iv) NNP_{FC} = GDP_{mp} − Consumption of fixed capital (depreciation) + Net factor income from abroad − Net indirect tax.

2. Income method

1. Compensation of employees

It may be defined as the sum total of all remuneration paid by an enterprise / employer to its employees in ration of their work.

Two components of COE :

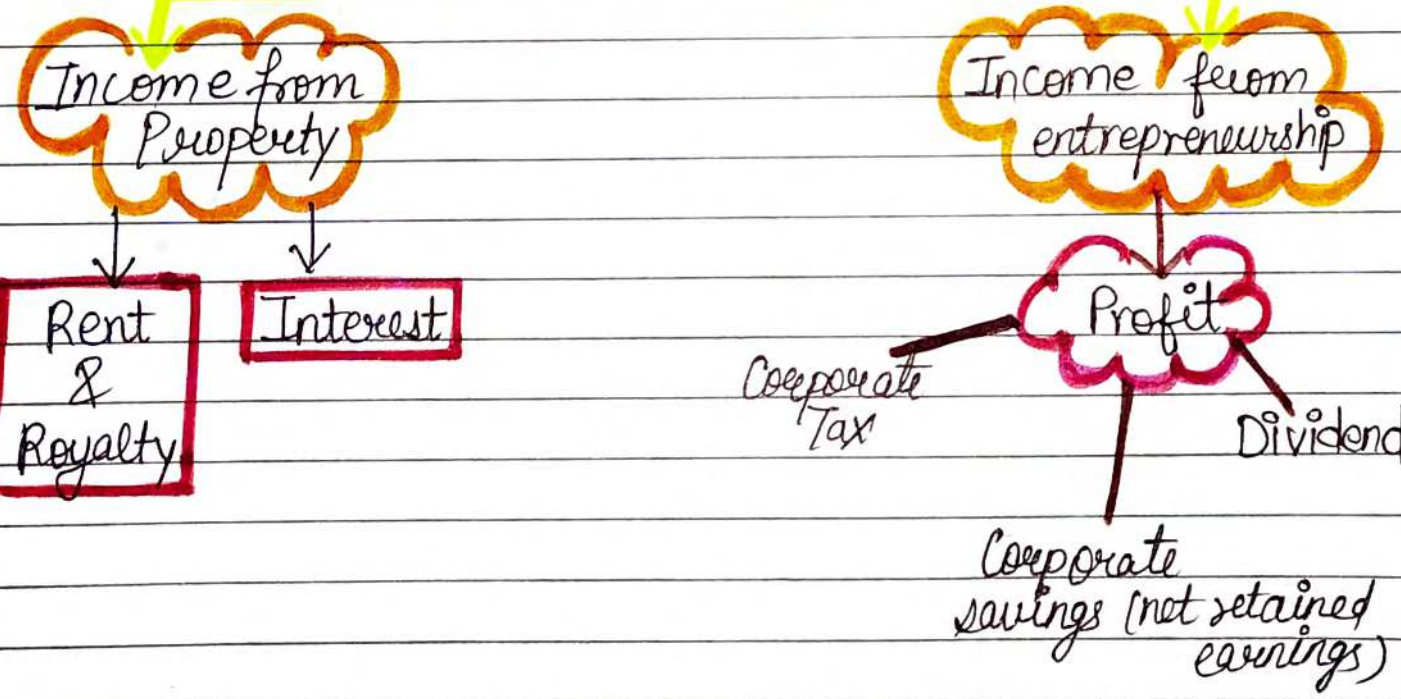
(i) Compensation of employees in cash
(ii) Compensation of employees in Kind

* COE in cash : It is the sum total of all remuneration in cash paid by an enter to rise to its employees in return of their work.

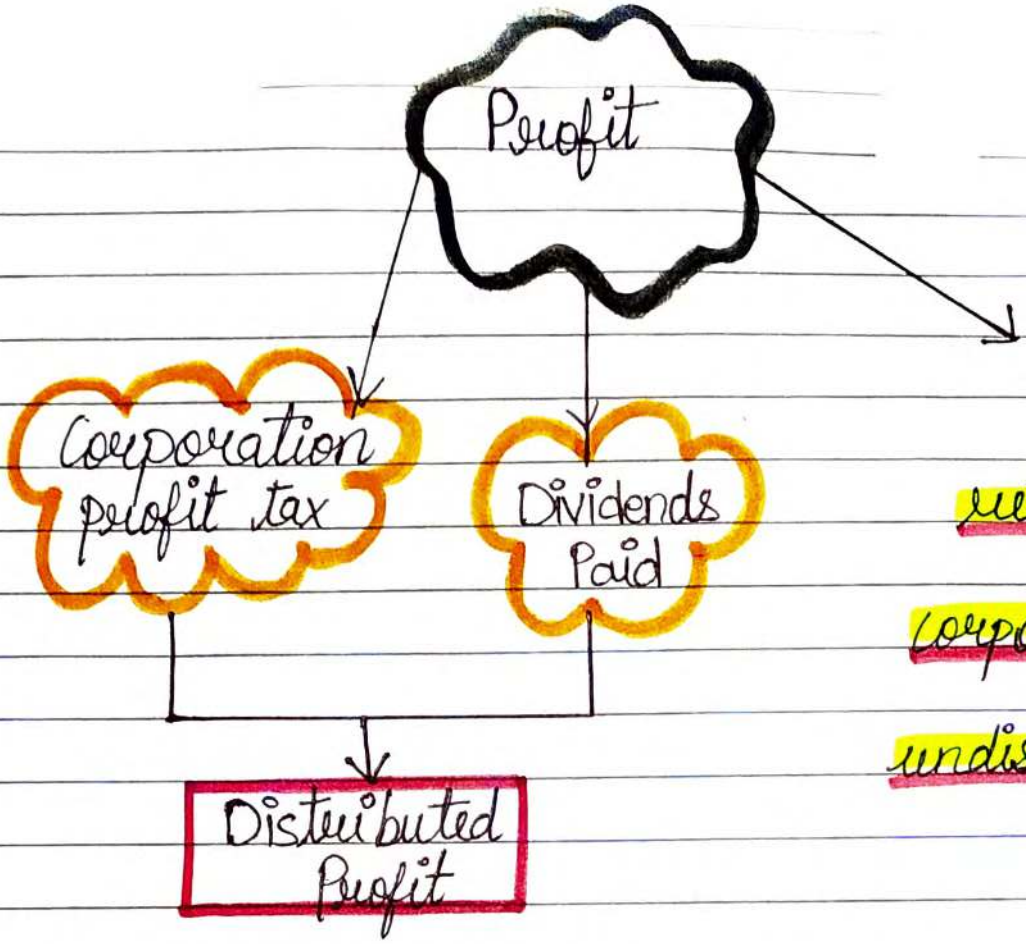
for example : Wages and salaries in cash bonus, employers contribution to social security scheme, allowance, pension etc.

* COE in kinds : It is the sum total of remuneration in kind paid by an enterprise to its employees in return of their work.
For example : wages and salaries in kind, rent free house, medical treatments, value of benefits provided in kind etc.

2. Operating Surplus



Profit को calculate करने की जरूरत Income method के through operating surplus निकालने के लिए होती है।



Calculation of Profit :

1. अगर Profit की value given है तो Profit को calculate करने की जरूरत नहीं होती है।
अगर Profit की Value, given नहीं है तो नीचे वाला कोई एक formula use होगा -

A Profit = Corporation Profit tax + dividends + undistributed Profit

= Distributed Profit + undistributed Profit
= Profit after tax + corporation profit tax
= Dividends + corporate income (corporate income = Profit tax + Related Earnings)

Mixed income of self-employed (MISE)

1. NDP_{FC} = (1) + (2) + (3) = COE + operating surplus + MISE
2. NNP_{FC} = NDP_{FC} + Net factor income from abroad.
3. GNP_{mp} = NDP_{FC} + consumption of fixed capital + Net Indirect tax.

3. Expenditure method

1. Government final consumption expenditure
2. Private final consumption expenditure
3. Net Export.
4. Gross domestic capital formation.

- Gross domestic fixed capital formation.
- Change in stock

GDP_{mp} = (1) + (2) + (3) + (4)

NNP_{FC} = GDP_{mp} - consumption of fixed capital + NFIA - NIT

Note: If capital formation is given as Net domestic capital formation we arrive at NDP_{mp}.

capital formation = Investment

→ Distinguish between stock and flow.

	Stock Variable	Flow Variable
Meaning	These are those variables whose quantity can be calculated at a particular point of times.	These are those variables whose quantity can be calculated over a period of time.
Nature	They are static in nature.	They are dynamic in nature.
Time dimension	They do not have time dimension	They have time dimension.
Examples	Balance sheet, wealth, National capital, money supply, opening stock, closing stock etc.	Capital formation, conversion, profit, loss, change in money supply, change in stock etc.

Gross domestic Product (GDP)

It may be defined as the money value of all goods and services produced within the domestic boundary of an economy during a period of time.

Limitations of GDP

The GDP is an index of the welfare of the people in the economy. But it is not an adequate measure of the welfare because of following limitations:

(1) Externalities
(2) Distribution of GDP or Inequality
(3) Composition of GDP
(4) Non-monetary Transactions
(5) Black money.

"GDP underestimate welfare" Justify—

It is true to say that GDP estimates welfare because it includes certain things which negatively affects the welfare. These things are stated below:

(1) GDP fails to include inequality.

2) Composition of GDP: GDP may rise due to rise in the production of those goods which negatively affects the welfare i.e liquor, tobacco. Products - gun, weapons etc.

3) GDP fails to include, Negative externalities like pollution, traffic Jams. It reduces the welfare.

→ Distinguish between Real GDP and Nominal GDP.

	Real GDP	Nominal GDP
1	It is the GDP, which is calculated at base year price level. i.e Real GDP = Base year price level X Quantity of output.	It is the GDP, which is calculated at current year price level. i.e Nominal GDP = current year price level x quantity of output.
2.	It changes due to change in physical output only.	It changes due to change in both i.e change in price level and change in output.
3.	It is better and more reliable index of welfare and growth.	Comparatively, it is not as good index of welfare and growth.

→ "GDP underestimate Welfare". Justify.

It is true to say that GDP underestimates welfare because it fails to include certain things which raises the welfare of the people in an economy. These things are stated below:

(1) GDP fails to include non-monetary transactions.
(2) GDP fails to include positive externalities like greenery, Knowledge as it raises the welfare.

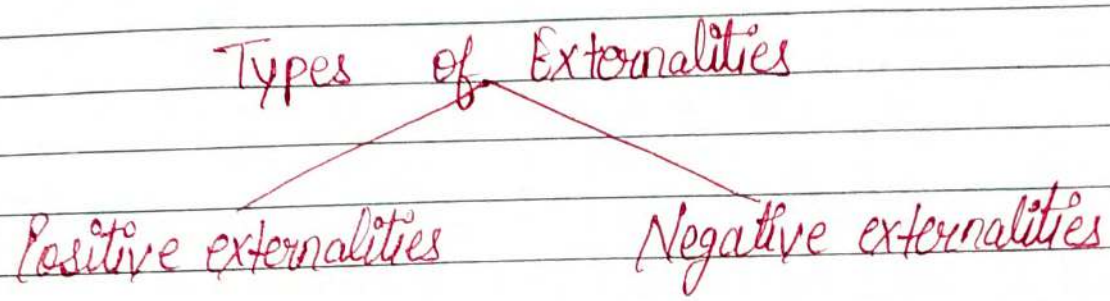
Real GDP Vs Nominal GDP

Real GDP is better index of the growth and welfare of an economy because real GDP changes due to change in only physical quantity and output. Hence, real GDP truly reflects the level of growth.

Defines :-

1 Externalities

It refers to good and bad impact of an activity. without paying the price or penalty for that.



2 Positive externalities.

It refers to good impact (benefit) of an activity without saying the price or penalty for that.
example: Greenery, Knowledge etc.
Impact of welfare: Positive externalities raises welfare of people in the economy i.e beautiful garden brings happiness and creates healthy environment.

3 Negative externalities

It refers to bad impact (harm) of an activity without paying price or penalty for that.
example: Pollution, traffic jam.
Impact of welfare: Negative externalities reduces the welfare of the people in the economy as it harms. for instance, pollution creates a lot of health Problems i.e. diseases.

4. **Real GDP**
It refers to GDP calculated at base year prices. It is also known as GDP at constant prices.

5. **Nominal GDP**
It refers to GDP calculated at current year prices. It is also known as GDP at current prices.

6. **GDP Deflator**
It refers to the ratio of Nominal GDP and Real GDP multiplied by 100.

$$\text{i.e. GDP Deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

7. **Real National Income**
It refers to national income calculated at base year prices.

8. **Nominal National Income**
It refers to national income calculated at current year prices.

9. **Income deflator** $= \frac{\text{Nominal Income}}{\text{Real Income}} \times 100$

10. **Real GNP**
It refers to GNP calculated at base year prices.

11. **Nominal GNP**
It refers to GNP calculated at current year prices.

12. GNP Deflator = $\frac{\text{Nominal GNP}}{\text{Real GNP}} \times 100$

The main steps involved in measuring national income through product method :

A classify the producing units into industrial sectors like primary, secondary and tertiary sectors.

B Estimate the net value added at the factor cost.

C Estimate the value of output by sales + change in stock.

D Estimate gross value added by value of output − intermediate cost.

E Deduct depreciation and net indirect tax from gross value added at market price to arrive at net value added at factor cost. = NDP_{FC}

F Add net factor income received from abroad to NDP_{FC} to obtain NNP_{FC} which is national income.

Steps involved in calculation of national income through income method :

A classify the producing enterprises into industrial sectors like primary, secondary and tertiary.

B Estimate the following factor income paid out by the producing units in each sector i.e

* Compensation of employees
* Operating surplus.
* Mixed income of self employed.

C Take the sum of the factor income by all the industrial sectors to arrive at the NDP_{FC} (which is called domestic income).

D Add net factor income from abroad to the net domestic product at factor cost to arrive at the net national product at factor cost.

Steps involved in calculation of national income through the expenditure method :

A Classify the economic units incurring final expenditure into distant groups like households, government, firms etc.

B Estimate the following expenditure on final products by all economic units -

* Private final consumption expenditure
* Government final consumption expenditure
* Gross domestic capital formation.
* Net export.

(Sum total of above gives GDPmp)

C Deduct depreciation, net indirect taxes to get NDP_{FC}

D Add net factor income from abroad to NDP_{FC} to arrive at NNP_{FC}

→ Precautions to be taken while calculating national income through product method:

(a) Avoid double counting of production, take only value added by each production unit.

b) The output produced for self-consumption to be included.

(c) The sale and purchase of second hand goods should not be included.

(d) Value of intermediate consumption should not be included.

(e) The value of services rendered in sales must be included.

→ Precautions to be taken while calculating national income through income method:

(a) Income from owner occupied house to be included.
(b) wages & salaries in cash and kind both to be included.
(c) Transfer income should not be included.
(d) Interest on loans taken for production only to be included. Interest on loan taken for consumption expenditure is non-factor income and so not included.

→ Precautions to be taken while calculating national income through expenditure method:

(a) Avoid double counting of expenditure by not including expenditure on intermediate product

(b) Transfer expenditure not to be included.

(c) Expenditure on purchase of second hand goods not to be included.

List of Items that should be included and should not be included in the calculation of National income, Domestic income, GDP, GNP etc.

Special case of "Interest"

Note: Basically interest is included as if it is a factor income but if purpose of loan is given in the statement the following treatment should be considered.

I) If interest is paid consumption purpose loan then it will not be included and it will be considered transfer payment.

(a) Interest on loan given by household / infinity to anybody (friend, Bank etc).

(b) National Debt interest or Interest on Public Debt.

(II) If interest is paid for production purpose loan then it will be included and it will be considered as factor payment.

(a) Interest of debenture
(b) Interest given by bank / firm / company anybody (individual / Bank / company) etc.

Special case for Domestic income and GDP, NDP....

For domestic Income (See the Place only)

If the earning has been done into the domestic boundary of the country then it will be included.

For example :-

(1) Profit earn by a branch of American bank in India → (included)
(2) Salary received by an Indian working in Russian embassy in New Delhi → (not included)
(3) Salary received by a Russian working in Indian Embassy in Russia (included)
(4) Profit earned by a branch of Indian bank in London. → (not included)

For National Income (See the person only)

If the earning has been done by the resident of the country then it will be included.

For example :-

(1) Profit earned by a branch of American bank in India. → (not included)

(2) Salary received by an Indian working in Russian Embassy in New Delhi → (included)

(3) Salary received by a Russian working in Indian Embassy in Russia. → (not included)

(4) Profit earned by a branch of Indian bank in London → (included)

Special case of Financial Assets

A Financial Assets are like shares, bonds, debentures etc.

B Any rise or fall in prices of financial Assets is not included as it is considered as CAPITAL GAIN.

I) Any sale/purchase of financial assets is not included because it is just a transfer of ownership title and it do not contribute to income flow.

II) Increase or Decrease in prices of financial Assets is not included as it is capital gain and do not contribute to income flow.

Remark: Brokerage commission on sale/purchase of financial assets is included as it is factor income.

(1) Transfer Income is not included in the calculation of National Income. For example -

(a) Gifts
(b) Charity
(c) Scholarship
(d) old-age pension
(e) Financial help given to flood victims
(f) Pocket money
g) money received by parents from their son working abroad
h) Prize won in a lottery (windfall gain)
i) Unemployment allowance
j) Claim received from insurance company by injured.
k) Taxes etc.

2.) Factor Income should be included in the calculation of National Income. For example -

a) Rent, interest, profit, wages, royals
b) Bonus
c) Gift received from employer
d) Pension
e) Brokerage fee (f) mixed income of self employed.

3. Intermediate expenditure should not be included in the calculation of National income. For example -

(a) Purchase of milk and vegetable by its.
(b) Purchase of chalks and dusters by schools.
(c) Expenditure on repair of machinery
(d) Expenditure on white wash of buildings.
(e) Expenditure on purchase of raw material.
(f) Fee paid to a lawyer or CA by product.
(g) Coal purchased by a factory.
(h) Soft drink purchased by school canteen.
(i) Electricity bill, Telephone bill, water
(j) Seeds, fertilizers, pesticides, purchase of a farmer
(K) Postage stamps purchased by a business.
(l) Machine purchased by a dealer of machine.
(m) expenditure on goods used in scientific research.

4. Final expenditure should be included in the calculation of National income. For example -

(a) Installation of machinery by a firm
(b) School fee paid by a student.
(c) Bus fare paid by a passenger.
(d) Anything purchased by household like clothes, vegetables, milk etc.
(e) Postage stamps purchased by a farmer
(f) Seeds purchased for kitchen gardening.
(g) Tractor purchased by a farmer.
(h) Sewing machine purchased by a school.
(i) Furniture purchased by a school.